

With IDAC1, the ADC measures the voltage across the RTD using the voltage across R_{REF} as the reference. This provides an output code that is proportional to the ratio of the RTD voltage and the reference voltage as shown in Equation 3. Ratiometric measurements will only produce positive output data, assuming zero offset error. For a fully-differential measurement, this is only the positive half of the full-scale range of the ADC, reducing the measurement resolution by one bit. The following equations assume a 24-bit bipolar ADC, with $\pm V_{REF}$ as the full-scale range of the ADC.

$$\text{Output code} = 2^{23} \cdot V_{RTD} / V_{REF} = 2^{23} \cdot I_{IDAC1} \cdot R_{RTD} / (I_{IDAC1} \cdot R_{REF}) \quad (3)$$

The currents cancel so that the equation reduces to Equation 4:

$$\text{Output code} = 2^{23} \cdot R_{RTD} / R_{REF} \quad (4)$$

In the end, the RTD resistance can be represented from the code as a function of the reference resistance.

$$R_{RTD} = \text{Output code} \cdot R_{REF} / 2^{23} \quad (5)$$

The measurement depends on the resistive value of the RTD and the reference resistor R_{REF} , but not on the IDAC1 current value. Therefore, the absolute accuracy and temperature drift of the excitation current does not matter. In a ratiometric measurement, as long as there is no current leakage from IDAC1 outside of this circuit, the measurement depends only on R_{RTD} and R_{REF} . ADC conversions do not need to be translated to voltage.

Assuming the ADC has a low gain error, R_{REF} is often the largest source of error. The reference resistor must be a high accuracy precision resistor with low drift. Any error in the reference resistance becomes a gain error in the measurement.

1.4.1 Lead Resistance Cancellation

In Figure 1-5, the lead resistances of a three-wire RTD are shown and a second excitation current source is added, labeled IDAC2.

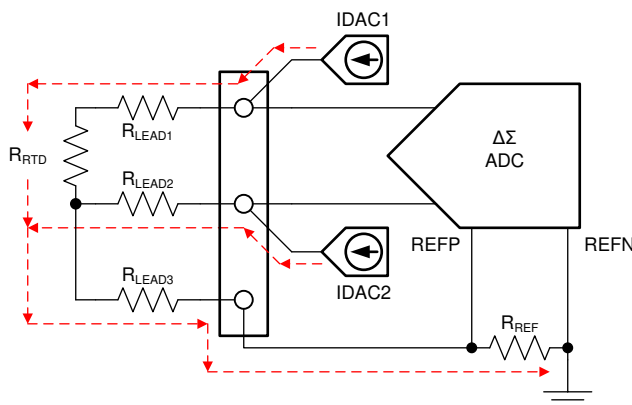


Figure 1-5. Example of Lead Wire Resistance Cancellation

With a single excitation current source, R_{LEAD1} adds an error to the measurement. By adding IDAC2, the second excitation current source is used to cancel out the error in the lead wire resistance. When adding the lead resistances and the second current source, the equation becomes:

$$\text{Output code} = 2^{23} \cdot [I_{IDAC1} \cdot (R_{RTD} + R_{LEAD1}) - (I_{IDAC2} \cdot R_{LEAD2})] / [(I_{IDAC1} + I_{IDAC2}) \cdot R_{REF}] \quad (6)$$

If the lead resistances match and the excitation currents match, then $R_{LEAD1} = R_{LEAD2}$ and $I_{IDAC1} = I_{IDAC2}$. The lead wire resistances cancel out so that Equation 6 reduces to the result in Equation 7 maintaining a ratiometric measurement.

$$\text{Output code} = 2^{23} \cdot R_{RTD} / (2 \cdot R_{REF}) = 2^{22} \cdot R_{RTD} / R_{REF} \quad (7)$$

$$R_{RTD} = \text{Output code} \cdot R_{REF} / 2^{22} \quad (8)$$